



Navigating the unknown: Leveraging self-information and diversity in partially observable environments

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ABSTRACT

Reinforcement learning algorithms often struggle to learn in partially observable environments, where different states of the environment may appear identical. However, not all partially observable environments pose the same level of difficulty for learning. This work introduces the concept of dissonance distance, a metric that can estimate the difficulty of learning in such environments. We demonstrate that self-information, such as internal oscillations or memory of previous actions, can increase the dissonance distance and make learning easier in partially observable environments. Additionally, sensory occlusion may occur after learning was completed, leading to a lack of sufficient information and catastrophic failure. To address this, we propose a spatially layered architecture (SLA) inspired by the brain, which trains multiple policies in parallel for the same task. SLA can change the amount of external information processed at each timestep, providing an adaptive approach to handle the changing information in the environment state-space. We evaluate the effectiveness of our SLA method showing learnability and robustness against realistic noise and occlusion in sensory inputs for the partially observable Continuous Mountain Car environment. We hypothesize that multi-policy approaches like SLA might explain the complex dopamine dynamics in the brain that cannot be explained with the state of the art scalar Temporal Difference error.

Complexity of learning in partially observable environments can be measured

Biology solves the issue of partial observation by including self-information, internal oscillations, which enable learning with partial information. We incorporate this mechanism into AI successfully.

Diversity in control policies is a recently described biological mechanism that enables robustness to sensory occlusion.

We introduce a spatially layered architecture that trains diverse policies in parallel for the same task and reaches great robustness against imperfect sensory information.

Reinforcement Learning, Partially Observable Environment, Control, Cognitive Model.

1. Introduction

Current artificial reinforcement learning (RL) agents are designed for predictable environments, where the environmental dynamics in which the agent learns and acts (may it be deterministic or stochastic) has fixed characteristics which are fully observable to the agent [1]. State-of-the-art RL agents often struggle and fail when lacking accurate predictions of the environment's dynamics [2]. This is in contrast to their claimed super-human performance in fixed environments [3,4] since their performance can catastrophically decline in the absence of complete information [5]. Partial observability is a major hurdle to

developing reliable fully autonomous agents.

In contrast, biological agents, such as humans, can adapt and act reasonably well even when facing sensory occlusion (such as occluding vision due to darkness or hearing due to competing noises). Likewise, the brain is capable of learning and adjusting to optimize rewards even in situations where the environmental changes cannot be fully predicted based on the available information [6]. The brain also optimizes the amount of energy expended depending on the predictability of states. Automatically reducing attention and compute in more predictable states while increasing attention in less predictable states [7]. Furthermore, recent work has demonstrated evidence that the brain learns a diverse set of policies at multiple timescales [8] and partly explains the complex dynamics of dopamine neurons that are not fully explained by a scalar TD-error. Similarly, dopamine ramps have been shown to predict “informativeness” of sensory inputs for future rewards, allowing modulation of the brain based on sensory feedback [9]. RL algorithms, on the other hand, utilize a scalar temporal difference (TD-error) that can only capture the difference in the expected reward and can not encode knowledge of the “informativeness” of the state.

This work aims to bridge the gap between RL and biological intelligence by presenting a bio-inspired solution. The main contributions of this work are as follows.

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1. Introducing “dissonance distance”, a new metric for assessing the difficulty of learning in a non-stationary environment using reinforcement learning.
2. Demonstrate that by adding internal time-keeping in the form of internal oscillators, RL algorithms can learn in non-stationary environments when the dissonance distance is relatively large. We also provide insights into how the brain might utilize self-information to increase dissonance distance.
3. Introducing the Spatially Layered Architecture (SLA), a bio-inspired framework for learning multiple policies in parallel (for the same environment) using information as an explicit constraint. This enables the agent to act well even when complete information is unavailable.

We open source our code at: <https://github.com/dee0512/SLA>.

2. Biological relevance

2.1. Layered architecture and policy diversity

The temporal difference (TD) error that reinforcement learning systems rely on has been used to explain the activity patterns of dopamine neurons in the brain [10–12]. However, the representations encoded by the dopamine neurons are far more complex than a simple scalar reward prediction error [13–21]. This is not surprising as scalar rewards are very inefficient and as a result, artificial RL algorithms require a lot of exploration. The problem is especially exacerbated when the rewards are delayed and sparse or the actions have multiple degrees of freedom as the agent now has to solve the difficult task of credit assignment for each scalar reward.

In our recent work, the temporally layered architecture (TLA) approach was used to create an agent that can learn and act at multiple timescales [22]. Similar to SLA which we introduce in this article, the TLA also trains each layer in parallel, where each layer trains using a different TD error but utilizing the same experiences. Multiple timesteps allow the agent to learn faster while reducing the amount of compute energy expended due to fewer decisions required. Parallel to TLA, recent work demonstrated the existence of multi-timescale reinforcement learning in the brain [8] that partly explains the complexity of the representations encoded by the dopamine neurons. However, we hypothesize that similar to multi-timescale reinforcement learning, the brain also employs multi-modal reinforcement learning similar to SLA. As we demonstrate in this paper, multi-modal RL is needed for robustness to sensory occlusion, learning in non-stationary environments, and energy conservation using the attention mechanism. A clue to this brain mechanism might lie in the existence of Dopamine ramps due to sensory feedback [9]. During any RL task, not all states (sensory inputs or feedback) have the same accuracy in estimating the value. As a result, the dopamine neurons capture the value and the “informativeness” of the sensory feedback. This results in the dopamine ramping up before the reward arrives, not only at learning time, but during processing, capturing the amount of information in the state rather than the TD-error (which should be 0 after learning). Analogous to this, in SLA, the switch networks activate when a state might be particularly informative to allow higher layers to process this information. Based on our results, we hypothesize that the dopamine ramps might not only capture the “informativeness” of the sensory feedback but also activate neural circuitry to utilize this information and modulate the attention towards it.

Additionally, since our work utilizes multiple-policies to perform the same task (based on different amounts of information), it is also related to ensemble learning [23–25]. Ensemble learning combines multiple models to improve predictive performance compared to individual models. In particular, active learning by query by committee methods are similar our approach [26–28]. This technique uses an ensemble of models (the committee) to determine which data points to query [29].

The committee members each vote on the labels, and the data points with the highest disagreement are selected for labeling, as these are considered the most informative. However, in our approach, all decisions are made by a single policy and the choice is between different policies that are based on different amounts of information which in turn affect the data distribution of experiences by their actions.

2.2. Spiking neural networks

Other works seek to solve the problems of artificial reinforcement learning by utilizing spiking neural networks (SNNs). SNNs, which mimic the discrete action potentials (spikes) of biological neurons, offer a biologically plausible framework for implementing RL algorithms. For example, SNNs have demonstrated the ability to learn from multiple time-scales of reward [30], thus paving the way to explain the complexity of dopamine activity. SNNs have also demonstrated superior robustness to noise and perturbations [30,31]. Thus, SNNs might be key in solving many of the challenges of RL discussed above, however, in this work, we focus on demonstrating the benefits of diverse policies and internal information for solving non-stationary and noisy environments. Therefore, we utilize artificial neurons, as they offer faster simulation speeds on traditional hardware.

2.3. Self-information and self-modeling

In this work, we demonstrate that self-information can be utilized to distinguish two states that are indistinguishable by their partial observations. Additionally, self-information also reduces reliance on external information, thereby improving the robustness to noise and adversarial attacks.

World models can also be considered as the ultimate form of self-information as they can model external environment and thus drastically reduce the reliance on external information. In model-based reinforcement learning, models are utilized for models are utilized for improving sample efficiency [32–34] and learning complex behaviors by leveraging a learned dynamics model to plan actions in imagination [35].

Similarly, there exists extensive literature on self-modeling in robotics and forward models in neuroscience. In neuroscience, forward models predict sensory outcomes of actions, aiding in motor control and adaptability [36–39]. In robotics, self-modeling involves creating internal representations for tasks such as motion planning and damage recovery [40]. These models enhance robot resilience and the ability to predict and imitate actions [41].

However, majority of previous works focus on modeling the external world and thereby reducing reliance on external information. However, such methods cannot model partially observable, non-stationary environments since the models are based on observations. In this work, we demonstrate how simple forms of self-information that is not strictly bound to the environment or the task at hand can be utilized to distinguish between two different states that look the same under partial observability. In future, this concept can be integrated with self-modeling and model-based RL to further enhance their capabilities.

3. The dissonance distance

In a reinforcement learning setting, an agent interacts with the environment using actions that result in changes in the environment, and the agent gaining rewards as a result of the actions and the changes. Reinforcement learning setups are typically defined as a 5-tuple $(\mathcal{S}, \mathcal{A}, p, R, \gamma)$. Where:

- $\mathcal{S}$: set of all possible states in the environment
- $\mathcal{A}$: set of all possible actions the agent can take

- p : Transition function that defines the probability of the next state given the current state and action. This is hidden from the agent. $p : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow [0, 1]$.
- R : Reward function. Typically hidden from the agent. $R : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$.
- γ : discount factor, balancing the benefit of current vs/future reward.

The goal of the agent is to maximize its reward during a single episode. Typically, the agent achieves this by assigning a value to each state or state-action pair. Reinforcement learning algorithms require stationary environments in order to learn successfully. That means, the transition function cannot be changed, ensuring that performing the same action in the same state will always have the same transition probability distribution. However, meeting this condition can be challenging, and handling non-stationary environments remains an open problem in reinforcement learning.

However, this constraint can be relaxed under certain conditions. We present a simple example in Fig. 1, where a two-dimensional grid world is considered. There is one positive and one negative reward state and the other states provide the 0 reward only. The agent has five available actions: Up, Down, Left, Right, and No-action. In this example, the agent cannot sense its vertical position (y-axis) and only has access to its horizontal position (x-axis). This limited information is sufficient for learning in the one-dimensional part of the environment on the right side. On the left side, the environment is non-stationary due to the partial information available to the agent. In this simple environment, the agent can learn to reach the reward state, since although the values assigned to the states on the left may be inaccurate, still, they will always be lower than those assigned to the stationary part (right) of the environment.

However, if the positive and negative rewards are flipped and the agent now needs to navigate to the left, the task becomes more difficult. This is because the agent encounters conflicting states like (2,1) and (2,2). In both these states, taking the action Left is rewarding. However, taking the action Up is rewarding positively to (2,1) and negatively to (2,2). It is worth noting that one can provide in advance optimal paths such as: (starting at state 4) Left, (state 3) Left, (state 2) Left, (state 1) Up, (state 1) Up. In this path, the states (1,0) and (1,1) have the same optimal action, and hence no dissonance exists along the path. However, making missteps may lead the agent to encounter conflicting states like (2,1) and (2,2). We define the dissonance distance as the minimum number of missteps from the optimal policy required to encounter conflicting states. In this case, the dissonance distance is 2, since we have to go from (2,0) to (2,1) and (2,2) to reach the conflicting states. The dissonance distance can be used as a measure of the difficulty of learning the task using random exploration, as it is correlated to the probability of encountering conflicting states. 1 is the worst possible dissonance, and an environment with higher dissonance will be easier to solve.

4. Result 1: Using oscillators to increase dissonance distance

Adding more information using sensors to make the environment stationary is often costly. To overcome this problem, the brain employs additional internal context or information to increase the dissonance distance. For example, brain waves or other internal oscillations can be used to augment the state space, adding a temporal context to each state [42]. This allows for distinguishing between states with the same sensory data visited at different times. Since time is often correlated with distance, it provides an easy way to increase the dissonance distance: Oscillators with period two, will manage to separate between odd and even time-steps. Oscillators with higher periods will provide a stronger separation among states.

To demonstrate this, we trained artificial agents with internal oscillations in a non-stationary environment. In the mountain car environment (Fig. 2), a car needs to reach a goal state outside a valley on the

right [44]. The car cannot simply accelerate and exit the valley due to insufficient power. Therefore, the optimal policy involves moving the car to the left first to build speed using gravity and then getting out of the valley. The stationary state space for this environment includes the car's position and velocity. However, if we remove the car's velocity from the state space, the problem becomes non-stationary and significantly more difficult, with a dissonance distance of approximately 2. This is because the car can move a single step to the left and then right to reach the same position with a different velocity. However, we can control the distance by augmenting the state space with oscillations which reduces the confusion as encountering two identical states will now be limited to the case where they also appear at the same time in the periodicity of the oscillator. The oscillator, while being independent to the environment, still provides information of the passage of time in the environment since time is fundamental to both the agent and the environment. The agent can leverage this to differentiate between two partially observable states that seem similar but differ in either their transition dynamics or reward.

We use the TD3 algorithm to train the agents in the OpenAI Gym implementation of the mountain car environment [43]. The TD3 algorithm is a state-of-the-art RL algorithm for training RL agents on continuous control tasks [45]. We augment the state-space with oscillators of different periods. The state of the oscillator is represented as a binary number. Thus, each bit of the oscillator input flips at a different frequency. Therefore, an n -bit oscillator will have n different bits flipping at different frequencies. The periodicity of an n -bit binary oscillator is thus 2^n . The results of our experiments are shown in table [Table 1]. We observe that the agent is unable to learn with no oscillators and for oscillation periods less than 256 ($n = 8$) (positive rewards indicate reaching the goal state). At the oscillation period of 256, the problem suddenly becomes learnable, despite the environment remaining non-stationary. At an oscillation period of 1024, the environment effectively becomes stationary, as the episode ends within 999 steps, ensuring that each state encountered by the agent is unique. However, confounding states still exist between episodes as the car starts at a different random location at the start of each episode. This experiment demonstrates the possibility of learning in non-stationary environments using a single internal oscillator. The amount of information employed can be adjusted to modify the dissonance distance, enabling a trade-off between problem difficulty and periodicity. Using other internal parameters to reduce the dissonance distance includes short-term memory of previous states, memory of previous actions, remaining energy levels, and more.

Table 1

Average reward of RL agent on non-stationary environments with binary oscillators of different bit lengths. As we increase the period of oscillations by increasing the bit-length, we also increase the distance between two confounding states thus making the problem solvable. All results are averaged over 5 trials and each trial was evaluated over 10 episodes.

Oscillator Bits (n)	Avg. Reward
No Oscillator	-0.001
1	-0.005
2	-0.007
3	-0.012
4	-0.02
5	-0.018
6	-0.077
7	-0.107
8	55.01
9	50.455
10	73.372

5. Result 2: Asymmetric actor-critic can learn control using partial information

Asymmetric actor-critic configurations can utilize information available only during training to create better critics for actors with partial information (Fig. 3). Previously, asymmetric actor-critic algorithms have been used to leverage full information available in simulators to train robots for real-world scenarios where only partial information is available [46]. Other approaches utilize history-based actors and state-based critics to tackle partial observability [47].

In this work, we utilize asymmetric actor-critic methods to reduce the amount of information required to operate and thus improve robustness to input noise. We assume that complete information is available during training and that the information available to the actor is a subset of the input to the critic. Our findings indicate that utilizing asymmetric actor-critic methods significantly improves performance under non-stationary information.

To test the efficacy of this approach, we experiment with the InvertedPendulum-v2 environment in the MuJoCo physics simulation [48]. We augmented the state-space with previous action and oscillations. Next, we trained a policy without the position and angle of the cart. When trained using a standard actor-critic setting, where both the actor and the critic lack the position and angle information, the agent failed to learn, never reaching a reward greater than 50. However, when trained using a critic with the privileged information of the angle and position, the agent found near-optimal (reward = 1000) policies, achieving an average reward and standard deviation of 967.16 ± 44.81 over 10 trials.

6. Spatially Layered Architecture

One remarkable property that the brain possesses is the ability to quickly adapt to changes in the information/sensing capabilities of the body. In contrast, artificial RL algorithms are brittle to any changes in the inputs.

Recent work has shown that the brain uses distributed control to allow multiple independent systems to process the environment and react accurately [49,50]. Different inputs are processed at different layers, each with connections to motor control, allowing the same muscles to be moved by different layers based on different information. Additionally, deeper layers in the brain that integrate different sensory modalities are engaged for more complex tasks [51].

Based on this insight, we present the Spatially Layered Architecture (SLA) (Fig. 4). The SLA trains layers of control policies in a hierarchical structure based on spatial information. Each layer trains a policy using increasing state-space dimensions, with the first layer having the least amount of information and the last layer having the most. In addition to the control policies, each layer (except the last) also includes a switch network which trains a simple binary policy based on the state and the action selected by the layer. The switch networks learn the information context or “informativeness” of each state and determine whether to execute the given action or proceed to the next layer for more information.

One might worry that since each incremental layer accepts more information, every layer will perform at least as well as the previous layers on complete information, and hence the switch networks will learn to always switch to the next layers, resulting in the last layer always being activated and the agent preferring to accept all information. To prevent this from happening, we introduce a constraint in the form of an information penalty.

In biology and practical applications, an energy cost is associated with activating sensory circuits and gathering more information. There is also a cost associated with processing the new information. As we will demonstrate in the next section, assuming a complete information can lead to over-reliance on external feedback or information, which can result in catastrophic failure when the information is occluded.

In our approach, deeper or higher layers will only be activated when there is a significant increase in performance, taking into account the relatively “constant” increase in energy. To control this trade-off, an information penalty (p) reduces the reward of each layer that chooses to refer to the next layer. For each layer $l \in L$, where L is the number of layers in the SLA, the reward is calculated by:

$$R_t^l = R_t - p \cdot g_t^l$$

where R_t^l is the augmented reward received by the layer l at time t , R_t is the reward received from the environment at time t and $g \in \{0, 1\}$ is the switch action so that $g = 1$ activates the next layer. The penalty p can be adjusted according to the application and can also differ from layer to layer. However, for this work, we use a constant $p = 0.01$.

To train all layers simultaneously, each layer observes the actions taken in the environment and the state of the environment at each timestep. This allows all layers to gather transitions as experiences to learn from. However, since each layer receives different state information, due to sensory differences, and potentially different reward due to the information penalty, each layer may learn a slightly different policy for the same task. Similarly, the switch networks also collect experience for each transition. In the switch network, if a layer is never activated, its corresponding switch action is recorded as 0. This is justified because if the layer is not activated, additional information is not necessary for that specific state, and therefore the higher layers should not be activated.

Furthermore, it is worth noting that the SLA approach may lead to interesting behaviors that conserve energy in the long term. This is because the information penalty is included in the reward, and each layer aims to optimize the total discounted reward rather than just the one-step reward. This aspect could be particularly valuable in understanding the differences between human and AI behaviors, as current AI algorithms do not consider the trade-off between the energy required to process current and future states and the reward from the environments [52].

7. Result 3: reducing sensory reliance via spatially layered architecture

To demonstrate the robustness gained by SLA, we evaluate it in the mountain car environment with a three-layer policy. The inputs to each layer are as follows.

4. Layer 1: Previous action + Internal Binary Clock
5. Layer 2: Layer 1 + Velocity of the Cart
6. Layer 3: Layer 2 + Position

As a comparison and ablation study, we trained three standalone policies using the TD3 algorithm with no internal information (TD3), the same information as Layer 2 (TD3 + Internal - POS) and the same as Layer 3 (TD3+internal), respectively. The TD3, TD3+Internal-POS, and TD3+Internal are trained in a standalone manner using all information equally. In contrast, the layers of SLA are trained using the information penalty approach described in Section [SLA] so that the agents focus on the internal information and use external information only if required. To evaluate the robustness of the policies to changes in the environment, we evaluated the policies on the mountain car environment where we varied the probability of the external information to be occluded. Fig. 5 shows the results of the experiments. The x-axis represents the probability of occlusion of the external information (Velocity + Position), with 0 meaning perfect information and 1 meaning no external information. All policies are evaluated over 10 trials and each trial is evaluated over 10 episodes. As expected, TD3, which does not contain any self-information has the worst level of resilience, demonstrating the value of internal information in robustness to sensory occlusion in addition to learning in non-stationary environments. On the other hand, relying on too much information also leads to lack of resilience: Layer 3 and

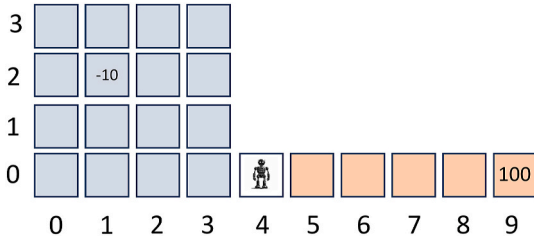


Fig. 1. Grid world environment where the agent can only observe its x-axis. The part of the environment on the left (blue) is therefore non-stationary for the agent while the part of the environment on the right (orange) is stationary. The numbers represent the reward gained by the agent upon reaching the state. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

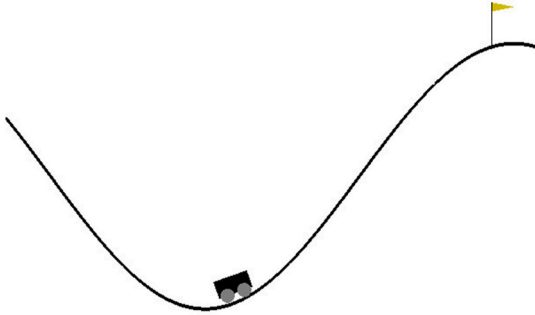


Fig. 2. Mountain Car Continuous environment [43]. The flag represents the goal state.

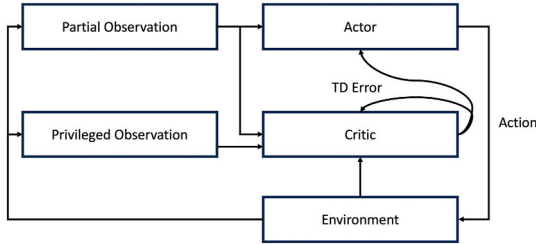


Fig. 3. Asymmetric actor-critic where the critic can access complete information to train an actor on partial state observation. Asymmetric actor-critic can significantly improve performance for policies based on partially observable information.

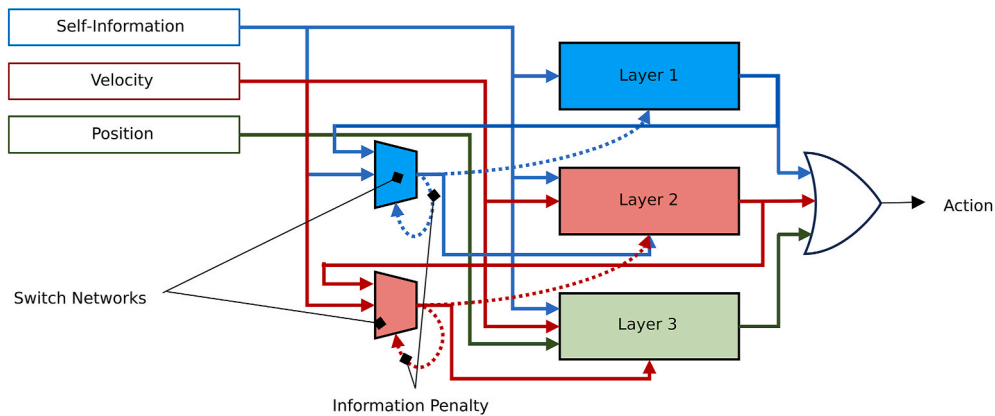


Fig. 4. A three-layered Spatially Layered Architecture (SLA). Each layer integrates incremental information. The switch networks are responsible for activating the layers and modulating the rewards to the layers using information penalty.

TD3+Internal, which rely on all possible information and expect the environment to be stationary, start failing quickly as the external information is occluded. The performance of Layer 1 is constant since it does not depend on external information. Significantly, Layer 2 is more robust to occlusion compared to the standalone policy (TD3+Internal-POS) demonstrating the value of the SLA architecture since both policies contain internal information. Finally, the SLA combines the benefits of all the layers, adjusting its policy online as the amount and quality of information changes.

Additionally we also present similar robustness results on the Inverted Pendulum benchmark. For Inverted Pendulum, we utilize a two-layered SLA.

7. Layer 1: Previous action + Internal Binary Clock + Velocity and Angular Velocity of the cart

8. Layer 2: Layer 1 + Position and Angle of the pole

The problem of the inverted pendulum with missing information is more difficult than that of the mountain car and cannot be solved using internal information. This results in very low performance for layer 1, which operates under conditions of missing information (Fig. 6). Additionally, layer 2 also shows significant decay in performance as the occlusion increases. However, surprisingly, the SLA, which switches between layer 1 and layer 2, demonstrates a superior emergent robustness that is not present in either layer alone. For example, at a 0.5 probability of occlusion, both layer 1 and layer 2 have an average reward of less than 50, yet the SLA has an average reward of more than 700. Lastly, similar to the previous environment, TD3 + internal demonstrates lower robustness despite being trained on the same information as layer 2.

We conclude that SLA constitutes a useful architecture for learning and operating in compromised environments with partial observability or damage to input sensors.

8. Discussion and concluding remarks

Non-stationary environments are common in the real world. For intelligent agents with limited resources like time and energy, learning to act and predict real-world dynamics is challenging. In biology, developing new senses to reduce non-stationarity in the environment comes at a high evolutionary cost. Even with access to information, there is a cost to processing it, which must be balanced against the value of potential rewards. For example, food is a common reward for biological agents, analogous to energy. The energy spent in obtaining food should be less than the energy gained from it, resulting in a net positive energy gain. Likewise, cognitive energy should only be used when necessary. The brain achieves this by utilizing different networks with

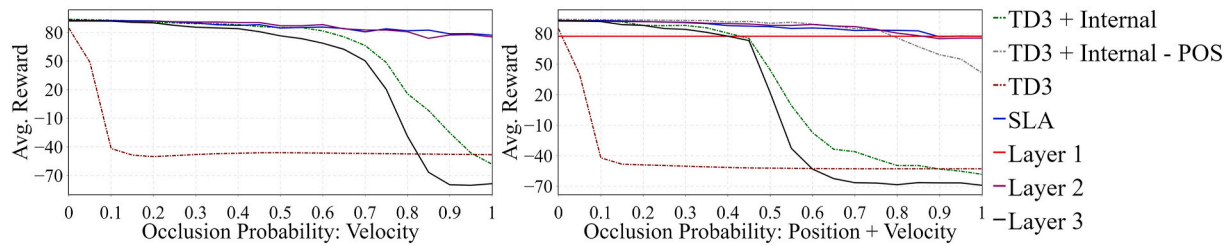


Fig. 5. A layered architecture combines the benefits of multiple policies, each trained to act based on different amounts of spatial information (Layers 1,2,3). Thus the agent can change its policy on the fly and adapt to missing information (SLA). In comparison, an agent that is trained on all the available information (TD3 + Internal) cannot adapt to missing information after training. We also see that a standalone policy (TD3 + Internal - POS) trained on the same information as layer 2 is less robust than layer 2; this is since the SLA layers use an information penalty which train them to use external information only when absolutely necessary. Finally, a policy that does not rely on internal information (TD3) has the worst performance. All results are averaged over 10 trials. The value 0 on the x-axis means no occlusion, or perfect information, while 1 on the x-axis means no information at all is received. Left: Results for velocity information occlusion. Right: Results for position + velocity information occlusion.

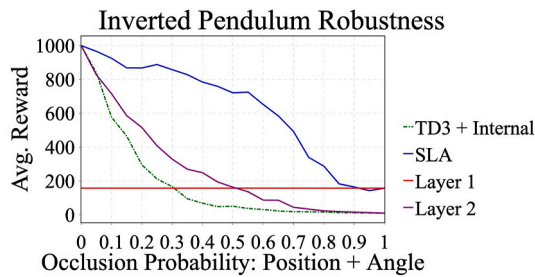


Fig. 6. SLA and TD3 performance on the difficult problem of inverted pendulum with varying probability of missing position and angle of the pole. SLA can intelligently switch between layer 1 and layer 2 policies to demonstrate emergent robustness not present when the layers are tested individually.

varying cognitive costs, sensory inputs, and speeds. By switching between these networks, the brain can trade-off energy, speed, and computation based on the situation. This forms the basis of adaptive reaction time and attention [53]. Interestingly, the ability to use as much precision as required and save on precision in computing as much as possible, is a hallmark of Super-Turing computation, leading to its superior computational capabilities over today's computers [54,55].

Artificial agents, in general, can react much faster than the brain and can use huge amounts of energy to process information quickly. And this indeed is the state of the art in AI, which has led to unprecedented energy spending in technology [56]. The artificial intelligence community has largely overlooked the benefits of the brain's diversity. In addition to energy waste there are other important reasons to follow the path of biological decision making. First, while a fast agent can react quickly, learning to react at extremely high speeds is very challenging due to the relatively slow changes and feedback from the environment [57]. Secondly, an agent with a large array of sensors can predict environmental dynamics, but a failure of any of them may lead to cascading failures. This is particularly critical for autonomous agents like self-driving cars and autonomous drones [58,59].

While there exists extensive previous work on dealing with non-stationary environments, almost all previous works try to "fix" the non-stationary environments, either by adding more information [60, 61] or splitting the environment into smaller stationary environments [62,63]. However, solutions that add more information often are unscalable once the agent is deployed in the environment as it requires more sensory inputs which may come at the expense of time to collect and fuse, additional energy spending, and reduction in robustness in realistic environments when not all inputs are reliable. Similarly, approaches that split the environment into smaller stationary environments require a lot of exploration for the identification of the sub-environments, making them inefficient and sometimes unsafe.

In contrast, we demonstrate the realistic importance of learning to act in partially observable and unpredictable environments, rather than attempting to increase the observability and predictability of the environments. We know this is possible since this is done regularly by animals. We first introduce the dissonance distance, a metric that measures the distance between conflicting states and the difficulty of learning in partially observable environments. We show that self-information, such as internal oscillations and previous action, can increase the dissolution distance and practically, which is often sufficient to solve partially observable environments. The internal oscillations in the brain, unlike our work, are not rigid and do not have a perfect time period. They change with time and also entrain based on the task at hand resulting in variational attention based on the oscillations [64,65]. In future, we hope to implement such oscillations to further improve on our current model.

We further introduce the Spatially Layered Architecture (SLA), a bio-inspired framework that learns a diverse set of policies in parallel, with varying amounts of information as input. The SLA uses information as a constraint, learning to reduce the amount of information required for any given situation. Thus, the SLA provides a cognitive model for spatial attention and demonstrates that spatial attention enables robustness in partially observable environments and when inputs are noisy or missing. Lastly, this work emphasizes the need for artificially intelligent agents to integrate constraints such as energy and information, even as we design larger and more energy-intensive AI systems.

CRediT authorship contribution statement

Devdhar Patel: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis. **Hava T. Siegelmann:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] R.S. Sutton, A.G. Barto, *Reinforcement Learning: an Introduction*, MIT press, 2018.
- [2] Papadimitriou C, Peng B. The Complexity of Non-stationary Reinforcement Learning. ArXiv. 2023;abs/2307.06877.
- [3] V. Mnih, K. Kavukcuoglu, D. Silver, A.A. Rusu, J. Veness, M.G. Bellemare, et al., Human-level control through deep reinforcement learning, *nature* 518 (7540) (2015) 529–533.
- [4] P.R. Wurman, S. Barrett, K. Kawamoto, J. MacGlashan, K. Subramanian, T. J. Walsh, et al., Outracing champion gran turismo drivers with deep reinforcement learning, *Nature* 602 (7896) (2022) 223–228.

- [5] L. Meng, R. Gorbet, D. Kulić, Partial Observability during Drl for Robot Control, 2022 arXiv preprint arXiv:220904999.
- [6] T.D. Ferguson, A. Fyshe, A. White, O.E. Krigolson, Humans adopt different exploration strategies depending on the environment, *Computational Brain & Behavior*. 6 (2023) 671–696.
- [7] A. Moors, J. De Houwer, Automaticity: a theoretical and conceptual analysis, *Psychol. Bull.* 132 (2) (2006) 297.
- [8] P. Masset, P. Tano, H.R. Kim, A.N. Malik, A. Pouget, N. Uchida, Multi-timescale reinforcement learning in the brain, *bioRxiv* (2023) 2023, 11.
- [9] J.G. Mikhael, H.R. Kim, N. Uchida, S.J. Gershman, The role of state uncertainty in the dynamics of dopamine, *Curr. Biol.* 32 (5) (2022) 1077–1087.
- [10] W. Schultz, P. Dayan, P.R. Montague, A neural substrate of prediction and reward, *Science*. 275 (5306) (1997) 1593–1599.
- [11] W. Schultz, Neuronal reward and decision signals: from theories to data, *Physiol. Rev.* 95 (3) (2015) 853–951.
- [12] J.Y. Cohen, S. Haesler, L. Vong, B.B. Lowell, N. Uchida, Neuron-type-specific signals for reward and punishment in the ventral tegmental area, *nature* 482 (7383) (2012) 85–88.
- [13] M.P. Gardner, G. Schoenbaum, S.J. Gershman, Rethinking dopamine as generalized prediction error, *Proceedings of the Royal Society B* 285 (1891) (2018) 20181645.
- [14] W. Dabney, Z. Kurth-Nelson, N. Uchida, C.K. Starkweather, D. Hassabis, R. Munos, et al., A distributional code for value in dopamine-based reinforcement learning, *Nature* 577 (7792) (2020) 671–675.
- [15] J. Cox, I.B. Witten, Striatal circuits for reward learning and decision-making, *Nat. Rev. Neurosci.* 20 (8) (2019) 482–494.
- [16] M. Watabe-Uchida, N. Uchida, Multiple dopamine systems: Weal and woe of dopamine, in: *Cold Spring Harbor Symposia on Quantitative Biology*, Cold Spring Harbor Laboratory Press, 2018, pp. 83–95.
- [17] B. Engelhard, J. Finkelstein, J. Cox, W. Fleming, H.J. Jang, S. Ornelas, et al., Specialized coding of sensory, motor and cognitive variables in vta dopamine neurons, *Nature* 570 (7762) (2019) 509–513.
- [18] S.J. Gershman, N. Uchida, Believing in dopamine, *Nat. Rev. Neurosci.* 20 (11) (2019) 703–714.
- [19] A.A. Hamid, J.R. Pettibone, O.S. Mabrouk, V.L. Hetrick, R. Schmidt, C.M. Vander Weele, et al., Mesolimbic dopamine signals the value of work, *Nat. Neurosci.* 19 (1) (2016) 117–126.
- [20] A. Mohebi, J.R. Pettibone, A.A. Hamid, J.M.T. Wong, L.T. Vinson, T. Patriarchi, et al., Dissociable dopamine dynamics for learning and motivation, *Nature* 570 (7759) (2019) 65–70.
- [21] J.D. Berke, What does dopamine mean? *Nat. Neurosci.* 21 (6) (2018) 787–793.
- [22] D. Patel, T. Sejnowski, H. Siegelmann, Temporally layered architecture for efficient continuous control, *arXiv preprint arXiv:230518701* (2023).
- [23] I.D. Mienye, Y. Sun, A survey of ensemble learning: concepts, algorithms, applications, and prospects, *IEEE Access* 10 (2022) 99129–99149.
- [24] J. Zhu, A. Zhang, H. Zheng, Research on predictive model based on ensemble learning. *Highlights in Science, Eng. Technol.* (2023).
- [25] J. Xu, P. Tan, J. Zhou, L. Luo, Online multi-task learning framework for ensemble forecasting, *IEEE Trans. Knowl. Data Eng.* 29 (2017) 1268–1280.
- [26] P. Melville, R. Mooney, Diverse ensembles for active learning, in: *Proceedings of the Twenty-First International Conference on Machine Learning*, 2004.
- [27] H. Hino, S. Eguchi, Active learning by query by committee with robust divergences, *Information Geometry* 6 (2022) 81–106.
- [28] M. Zhi-chun, Research of query-by-committee method of active learning and application, *Comput. Eng.* (2006).
- [29] H. Seung, M. Oppor, H. Sompolinsky, Query by Committee, 1992, pp. 287–294.
- [30] H. Anwar, S. Caby, S. Dura-Bernal, D. D’Onofrio, D. Hasegan, M. Deible, et al., Training a spiking neuronal network model of visual-motor cortex to play a virtual racket-ball game using reinforcement learning, *PLoS One* 17 (5) (2022) e0265808.
- [31] D. Patel, H. Hazan, D.J. Saunders, H.T. Siegelmann, R. Kozma, Improved robustness of reinforcement learning policies upon conversion to spiking neuronal network platforms applied to atari breakout game, *Neural Network.* 120 (2019) 108–115.
- [32] A.S. Anand, J.E. Kveen, F. Abu-Dakka, E.I. Grøtli, J.T. Gravidahl, Addressing sample efficiency and model-bias in model-based reinforcement learning, in: *2022 21st IEEE International Conference on Machine Learning and Applications (Icmla)*, IEEE, 2022, pp. 1–6.
- [33] D. Zhao, J. Liu, R. Wu, D. Cheng, X. Tang, Optimistic sampling strategy for data-efficient reinforcement learning, *IEEE Access* 7 (2019) 55763–55769.
- [34] Wang Z, Wang J, Zhou Q, Li B, Li H. Sample-efficient Reinforcement Learning via Conservative Model-Based Actor-Critic. *ArXiv*. 2021;abs/2112.10504.
- [35] Shi L, Lim JJ, Lee Y. Skill-based Model-Based Reinforcement Learning. *ArXiv*. 2022;abs/2207.07560.
- [36] R. Miall, D. Wolpert, Forward models for physiological motor control, *Neural Network. : the official journal of the International Neural Network Society* 9 (8) (1996) 1265–1279.
- [37] K.J. Friston, Learning and inference in the brain, *Neural Network. : the official journal of the International Neural Network Society* 16 (9) (2003) 1325–1352.
- [38] B. Webb, Neural mechanisms for prediction: do insects have forward models? *Trends Neurosci.* 27 (2004) 278–282.
- [39] H. Tanaka, T. Ishikawa, J. Lee, S. Kakei, The cerebro-cerebellum as a locus of forward model: a review, *Front. Syst. Neurosci.* 14 (2020).
- [40] B. Chen, R. Kwiatkowski, C. Vondrick, H. Lipson, Fully body visual self-modeling of robot morphologies, *Sci. Robot.* 7 68 (2022) eabn1944.
- [41] T. Ogata, R. Yokoya, J. Tani, K. Komatani, H.G. Okuno, Prediction and imitation of other’s motions by reusing own forward-inverse model in robots, in: *2009 IEEE International Conference on Robotics and Automation*, 2009, pp. 4144–4149.
- [42] L.G. Allan, The perception of time, *Percept. Psychophys.* 26 (5) (1979) 340–354.
- [43] G. Brockman, V. Cheung, L. Pettersson, J. Schneider, J. Schulman, J. Tang, et al., Openai Gym. *arXiv Preprint arXiv:160601540*, 2016.
- [44] A.W. Moore, Efficient Memory-Based Learning for Robot Control, University of Cambridge, 1990.
- [45] S. Fujimoto, H. Hoof, D. Meger, Addressing function approximation error in actor-critic methods, in: *International Conference on Machine Learning*, PMLR, 2018, pp. 1587–1596.
- [46] L. Pinto, M. Andrychowicz, P. Welinder, W. Zaremba, P. Abbeel, Asymmetric actor critic for image-based robot learning, *arXiv preprint arXiv:171006542* (2017).
- [47] A. Baisero, C. Amato, Unbiased asymmetric reinforcement learning under partial observability, *arXiv preprint arXiv:210511674* (2021).
- [48] E. Todorov, T. Erez, Y. Tassa, MuJoCo: a physics engine for model-based control, in: *2012 IEEE/rsj International Conference on Intelligent Robots and Systems*, IEEE, 2012, pp. 5026–5033.
- [49] Y. Nakahira, Q. Liu, T.J. Sejnowski, J.C. Doyle, Diversity-enabled sweet spots in layered architectures and speed–accuracy trade-offs in sensorimotor control, *Proc. Natl. Acad. Sci. USA* 118 (22) (2021) e1916367118.
- [50] P. Masset, P. Tano, H.R. Kim, A.N. Malik, A. Pouget, N. Uchida, Multi-timescale Reinforcement Learning in the Brain, *bioRxiv.*, 2023.
- [51] P. Taylor, J.N. Hobbs, J. Burroni, H.T. Siegelmann, The global landscape of cognition: hierarchical aggregation as an organizational principle of human cortical networks and functions, *Sci. Rep.* 5 (1) (2015) 18112.
- [52] S. Milani, A. Juliani, I. Momennejad, R. Georgescu, J. Rzepecki, A. Shaw, et al., Navigates like me: understanding how people evaluate human-like ai in video games, in: *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, 2023, pp. 1–18.
- [53] T. Popov, B. Gips, N. Weisz, O. Jensen, Brain areas associated with visual spatial attention display topographic organization during auditory spatial attention, *Cerebr. Cortex* 33 (7) (2023) 3478–3489.
- [54] H.T. Siegelmann, Neural Networks and Analog Computation: beyond the Turing Limit, Springer Science & Business Media, 2012.
- [55] E.D. Sontag, H. Siegelmann, On the Computational Power of Neural Nets, vol. 50, *J Comp Syst Sci.*, 1995, pp. 132–150.
- [56] R. Desislavov, F. Martínez-Plumed, J. Hernández-Orallo, Compute and Energy Consumption Trends in Deep Learning Inference, 2021 *arXiv preprint arXiv: 210905472*.
- [57] A. McGovern, R.S. Sutton, A.H. Fagg, Roles of Macro-Actions in Accelerating Reinforcement Learning, 1997.
- [58] J.R.V. Rivero, I. Tahiraj, O. Schubert, C. Glassl, B. Buschardt, M. Berk, et al., Characterization and simulation of the effect of road dirt on the performance of a laser scanner, in: *2017 IEEE 20th International Conference on Intelligent Transportation Systems (Itsc)*, IEEE, 2017, pp. 1–6.
- [59] M. Trierweiler, P. Caldelas, G. Gröninger, T. Peterseim, C. Neumann, Influence of sensor blockage on automotive lidar systems, in: *2019 IEEE Sensors*, IEEE, 2019, pp. 1–4.
- [60] R.T. Icarte, T.Q. Klassen, R. Valenzano, S.A. McIlraith, Reward machines: exploiting reward function structure in reinforcement learning, *J. Artif. Intell. Res.* 73 (2022) 173–208.
- [61] S. Padakandla, A survey of reinforcement learning algorithms for dynamically varying environments, *ACM Comput. Surv.* 54 (6) (2021) 1–25.
- [62] B.C. Da Silva, E.W. Basso, A.L. Bazzan, P.M. Engel, Dealing with non-stationary environments using context detection, in: *Proceedings of the 23rd International Conference on Machine Learning*, 2006, pp. 217–224.
- [63] E. Hadoux, A. Beynier, P. Weng, Sequential decision-making under non-stationary environments via sequential change-point detection, in: *Learning over Multiple Contexts (Lmce)*, 2014.
- [64] C.E. Schroeder, P. Lakatos, Low-frequency neuronal oscillations as instruments of sensory selection, *Trends in neurosciences* 32 (1) (2009) 9–18.
- [65] P. Lakatos, G. Karmos, A.D. Mehta, I. Ulbert, C.E. Schroeder, Entrainment of neuronal oscillations as a mechanism of attentional selection, *science*. 320 (5872) (2008) 110–113.